

$$\text{Let, } A^{in} = \begin{bmatrix} 0 & 1 & \dots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

$$\text{Cell number, } N = \begin{bmatrix} n_1 \\ n_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1} [1 \quad 1 \quad \dots]_{1 \times N_{mets}} = \begin{bmatrix} n_1 & n_1 & \dots \\ n_2 & n_2 & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}_{N_{CT}, N_{mets}}$$

$$\text{Consumption matrix, } C = A^{in} \odot N = \begin{bmatrix} 0 & C_{1,2} & \dots \\ \vdots & \ddots & \dots \\ C_{N_{CT},1} & C_{N_{CT},2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}, \text{ where } C_{i,j} = A_{i,j}^{in} n_i$$

$$\text{Let, } \tau = \text{Column sum}(C) = [1 \quad 1 \quad \dots]_{1 \times N_{CT}} C_{N_{CT} \times N_{mets}} = [T_1 \quad T_2 \quad \dots]_{1 \times N_{mets}}, \text{ where } T_j = \sum_{i=1}^{N_{CT}} C_{i,j}$$

$$\text{and } \tau' = \begin{bmatrix} 1 \\ 1 \\ \vdots \end{bmatrix}_{N_{CT} \times 1} \begin{bmatrix} \frac{1}{T_1} & \frac{1}{T_2} & \dots \end{bmatrix}_{1 \times N_{mets}} = \begin{bmatrix} \frac{1}{T_1} & \frac{1}{T_2} & \dots \\ \frac{1}{T_1} & \frac{1}{T_2} & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}_{N_{CT} \times N_{mets}} \quad \forall T_i > 0, \text{ and } 0 \text{ otherwise}$$

Normalised consumption matrix, $C_{norm} = C \odot \tau'$

$$= \begin{bmatrix} 0 & C_{1,2} & \dots \\ \vdots & \ddots & \dots \\ C_{N_{CT},1} & C_{N_{CT},2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \odot \begin{bmatrix} \frac{1}{T_1} & \frac{1}{T_2} & \dots \\ \frac{1}{T_1} & \frac{1}{T_2} & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}_{N_{CT} \times N_{mets}} = \begin{bmatrix} 0 & \frac{C_{1,2}}{T_2} & \dots \\ \vdots & \ddots & \dots \\ \frac{C_{N_{CT},1}}{T_1} & \frac{C_{N_{CT},2}}{T_2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

Let Diet, $D = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$

Net Intake, $I = C_{norm} D$

$$= \begin{bmatrix} 0 & \frac{C_{1,2}}{T_2} & \dots \\ \vdots & \ddots & \dots \\ \frac{C_{N_{CT},1}}{T_1} & \frac{C_{N_{CT},2}}{T_2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

$$= \begin{bmatrix} I_1 \\ I_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}, \text{ where } I_k = \sum_{j=1}^{N_{mets}} \frac{C_{k,j}}{T_j} D_j$$

Let, $A^{out} = \begin{bmatrix} 0 & 1 & \dots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}$ and Byproduct fraction, $f = 0.5$

For each celltype, Out degree, $D^{out} = \text{Row sum}(A^{out}) = A^{out} \begin{bmatrix} 1 \\ 1 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$

$= \begin{bmatrix} D_1^{out} \\ D_2^{out} \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$, where $D_i^{out} = \sum_{j=1}^{N_{mets}} A_{i,j}^{out}$

Let, inverse out-degree, $\Omega = \begin{bmatrix} \frac{1}{D_1^{out}} \\ \frac{1}{D_2^{out}} \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$

Output multiplier, $O = f \left(\Omega_{N_{CT} \times 1} \begin{bmatrix} 1 & 1 & \dots \end{bmatrix}_{1 \times N_{mets}} \right) = \begin{bmatrix} \frac{f}{D_1^{out}} & \frac{f}{D_1^{out}} & \dots \\ \frac{f}{D_2^{out}} & \frac{f}{D_2^{out}} & \dots \\ \vdots & \vdots & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}$

Output, $S = A^{out} \odot O$

$$= \begin{bmatrix} S_{1,1} & S_{1,2} & \cdots \\ S_{2,1} & S_{2,2} & \cdots \\ \vdots & \vdots & \cdots \end{bmatrix}_{N_{CT} \times N_{mets}}, \text{ where } S_{i,j} = \frac{f}{D_i^{out}} A_{i,j}^{out}$$

Secreted metabolome part, $M^s = S^T I$

$$= \begin{bmatrix} S_{1,1} & S_{2,1} & \cdots \\ S_{1,2} & S_{2,2} & \cdots \\ \vdots & \vdots & \cdots \end{bmatrix}_{N_{mets} \times N_{CT}} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$$

$$= \begin{bmatrix} \sum_{k=1}^{N_{CT}} S_{k,1} I_k \\ \sum_{k=1}^{N_{CT}} S_{k,2} I_k \\ \vdots \end{bmatrix}_{N_{mets} \times 1} = \begin{bmatrix} \sum_{k=1}^{N_{CT}} I_k \frac{f}{D_k^{out}} \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

For each metabolite, In-degree, $D^{in} = \text{Column sum}(A^{in})$

$$= [1 \quad 1 \quad \dots]_{1 \times N_{CT}} A^{in} = [D_1^{in} \quad D_2^{in} \quad \dots]_{1 \times N_{mets}}, \text{ where } D_i^{in} = \sum_{j=1}^{N_{CT}} A_{i,j}^{in}$$

Net predicted metabolome, $M^{pred} = M^{unused, T} + M^s$,

$$\text{where } M_k^{unused} = D_k \quad \forall \quad D_k^{in} = 0$$

and 0 otherwise

Toy example

Let, $A^{in} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}_{2 \times 3}$ and Cell number, $N = \begin{bmatrix} 100 & 100 & 100 \\ 200 & 200 & 200 \end{bmatrix}_{2 \times 3}$

Consumption matrix, $C = A^{in} \odot N = \begin{bmatrix} 0 & 100 & 100 \\ 0 & 0 & 200 \end{bmatrix}_{2 \times 3}$

Let, $\tau = \text{Column sum}(C) = [0 \quad 100 \quad 300]_{1 \times 3}$ and $\tau' = \begin{bmatrix} 0 & \frac{1}{100} & \frac{1}{300} \\ 0 & \frac{1}{100} & \frac{1}{300} \end{bmatrix}_{2 \times 3}$

Normalised consumption matrix, $C_{norm} = C \odot \tau' = \begin{bmatrix} 0 & 100 & 100 \\ 0 & 0 & 200 \end{bmatrix}_{2 \times 3} \odot \begin{bmatrix} 0 & \frac{1}{100} & \frac{1}{300} \\ 0 & \frac{1}{100} & \frac{1}{300} \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 0 & 1 & \frac{1}{3} \\ 0 & 0 & \frac{2}{3} \end{bmatrix}_{2 \times 3}$

Let Diet, $D = \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}_{3 \times 1}$

Net Intake, $I = C_{norm} D = \begin{bmatrix} 0 & 1 & \frac{1}{3} \\ 0 & 0 & \frac{2}{3} \end{bmatrix}_{2 \times 3} \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 28.33 \\ 16.67 \end{bmatrix}_{2 \times 1}$

$$A^{out} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}_{2 \times 3} \quad \text{Inverse out-degree, } \Omega = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{1} \end{bmatrix}_{2 \times 1}$$

Byproduct fraction, $f = 0.5$

$$\text{Output multiplier, } O = \begin{bmatrix} \frac{0.5}{2} & \frac{0.5}{2} & \frac{0.5}{2} \\ 0.5 & 0.5 & 0.5 \end{bmatrix}_{2 \times 3}$$

$$\text{Output, } S = A^{out} \odot O$$

$$= \begin{bmatrix} \frac{0.5}{2} & \frac{0.5}{2} & 0 \\ 0.5 & 0 & 0 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 0.25 & 0.25 & 0 \\ 0.5 & 0 & 0 \end{bmatrix}_{2 \times 3}$$

$$\text{Secreted metabolome part, } M^s = S^T I = \begin{bmatrix} 0.25 & 0.5 \\ 0.25 & 0 \\ 0 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 28.33 \\ 16.67 \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} (0.25 \times 28.33) + (0.5 \times 16.67) \\ (0.25 \times 28.33) + (0 \times 16.67) \\ 0 + 0 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 15.4175 \\ 7.0825 \\ 0 \end{bmatrix}_{3 \times 1}$$

$$M^{unused} = [10 \quad 0 \quad 0]_{1 \times 3}$$

Net predicted metabolome, $M^{pred} = M^{unused, T} + M^s$,

$$= \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1} + \begin{bmatrix} 15.4175 \\ 7.0825 \\ 0 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 25.4175 \\ 7.0825 \\ 0 \end{bmatrix}_{3 \times 1}$$